

Introduction to Machine Learning

& Data Science Workflows

Applied Machine Learning — Session 1, Chapter 1

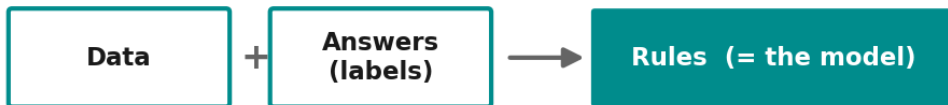
What Is Machine Learning?

Traditional programming



The computer writes the rules — we supply examples.

Machine learning



"A computer program is said to learn from experience E with respect to some task T and performance measure P , if its performance at T , as measured by P , improves with experience E ." — Tom Mitchell, 1997

Why Machine Learning?

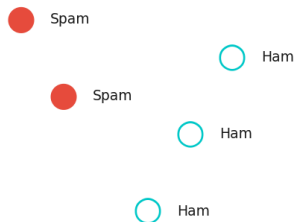
- Email spam → millions of patterns, impossible to hand-code
- Image recognition → pixels to meaning
- Medical diagnosis → beyond human attention span
- Recommendations → individual preferences at scale

Key insight: When rules are too complex to write — let data write them.

Three Paradigms of ML

Three Paradigms of Machine Learning

Supervised Learning
 $(X, y) \rightarrow \text{predict } y$



Labels provided $\rightarrow$ Model learns boundary

Unsupervised Learning
 $X \text{ only} \rightarrow \text{find structure}$



No labels $\rightarrow$ Algorithm discovers groups

Reinforcement Learning
 $\text{trial \& error} \rightarrow \text{max reward}$



Goal: maximize cumulative reward

Supervised Learning

- Training data has **labels**
- Model learns: $X \rightarrow y$
- Two flavors:
 - **Regression** $\rightarrow$ continuous output (price, temperature)
 - **Classification** $\rightarrow$ discrete output (spam/not spam, disease/no disease)

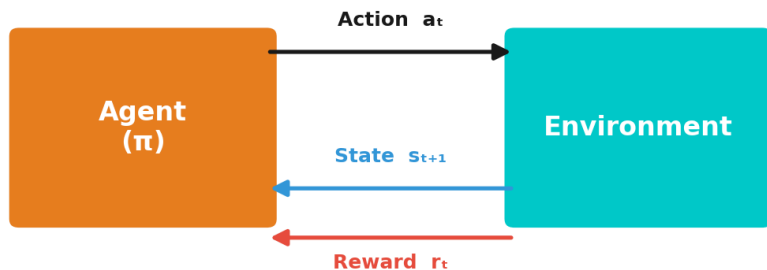
Unsupervised Learning

- **No labels** — only raw data
- Model finds hidden structure
- Applications:
 - Customer segmentation
 - Anomaly detection
 - Dimensionality reduction
 - Topic modeling

Reinforcement Learning

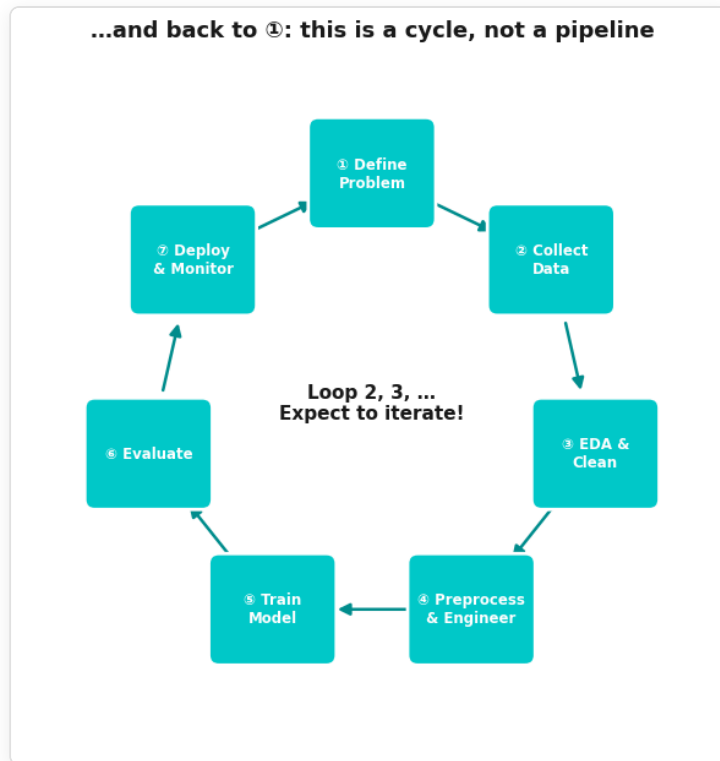
- **Agent** interacts with **Environment**
- Takes **Actions** → receives **Rewards**
- Goal: maximize cumulative reward

The Reinforcement Learning Loop



Goal: maximize total cumulative reward

The Data Science Workflow



Step ① — Define the Problem

- What are we trying to predict/discover?
- Who uses the result?
- What is a "good enough" answer?

Most ML project failures happen here, not in the model.

Steps ②③④ — Data, EDA & Preprocessing

- Where does the data come from? Is it representative? Legal/ethical constraints?
- Explore distributions, relationships, outliers
- Handle missing values, encode categories, scale features
- Split: train / test (validation → later)

Garbage in → garbage out. ~80% of project time lives here.

Steps ⑤⑥⑦ — Train, Evaluate, Deploy

- Choose algorithm(s), fit on training data
- Measure performance on **held-out** data — always compare with a simple **baseline**
- A model in a notebook helps no one → deploy, monitor drift, retrain

The cycle is the method. Expect at least 3 full loops.

The Python ML Ecosystem

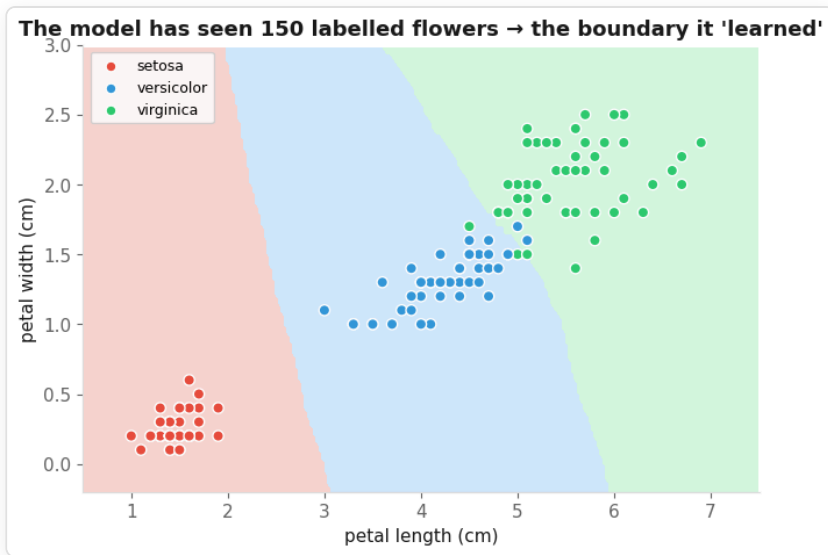
Tool	Role
<code>numpy</code>	Arrays & math
<code>pandas</code>	Data wrangling
<code>matplotlib</code> / <code>seaborn</code>	Visualization
<code>scikit-learn</code>	Algorithms & pipelines
<code>gymnasium</code>	RL environments

Now: Live Demo!

→ Open `ch01_introduction_examples.ipynb`

We will:

1. Load the famous **Iris dataset**
2. Inspect its structure
3. Visualize features → **see the pattern before any model**
4. Sneak preview: a baseline and a first model



Quick Check

Which paradigm is it?

1. Predicting tomorrow's electricity demand from the last 5 years
2. Grouping news articles into topics nobody defined beforehand
3. A thermostat that learns to keep you comfortable from your reactions
4. Deciding whether a credit card transaction is fraud

Quick Check

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Answers: 1 → Supervised (regression) · 2 → Unsupervised (clustering) · 3 → Reinforcement · 4 → Supervised (classification)

Key Takeaways

- ML = learning patterns from data automatically
- Three paradigms: supervised / unsupervised / reinforcement
- Workflow is a **cycle** — expect iteration
- Great ML starts with **understanding the data**
- Judge models on **unseen data**, against a **baseline**

Next: Chapter 2

Data Selection, Cleaning & Preparing

"Real-world data is messy, incomplete, and full of surprises. Let's learn how to tame it."